

B. Tech Degree VIII Semester Examination, April 2010**ME 802 MACHINE DESIGN II***(2002 Scheme)*

Time : 3 Hours

Maximum Marks : 100

(Use of design data book is permitted)
(Assume reasonable data wherever applicable)

- I. a. Select a suitable chain drive to transmit 30KW from an electric motor to a line shaft. Motor shaft diameter is 60mm, speed of motor shaft is 1200 rpm and that of line shaft is 250 rpm. Centre distance is 600 mm. (15)
b. Design a Cone clutch with leather facing cast iron mating surface to transmit 15KW at 750 rpm. The mean diameter is limited to 250mm and slope of the face is 15° . (10)
- OR**
- II. a. A multiplate clutch is to be designed to transmit 50KW at 500 rpm. Design the clutch. (20)
b. Define (i) Self energizing brake (ii) Self locking brake. (5)
- III. a. A cast steel pinion rotating at 900 rpm is to drive a cast iron gear at 144 rpm. The teeth are to have standard 20° stub involute profiles and maximum power to be transmitted is 25KW. Design the gear and pinion from the stand point of strength, dynamic strength and wear. Assume that the pinion is surface hardened to B.H.N. of 250 and gear to B.H.N. of 200. (17)
b. A pair of parallel helical gears of 18 teeth pinion meshing with 45 teeth gear. A 7.5 KW power at 2000 rpm is supplied to pinion. The normal module is 6mm. The helix angle is 23° and pressure angle is 20° . Determine tangential, radial and axial components of tooth force. (8)
- OR**
- IV. a. A pair of bevel gears is required to transmit 15 H.P. at 500 rpm, from driver shaft to driven shaft. The speed reduction is to be 3:1. The shafts are inclined at 60° . Pinion is to be made of cast steel and gear of cast iron. The service factor is 1.15. Design the gears? (18)
b. Explain about the advantages of gears over other drives. (7)
- V. a. Derive petroff's equation as applied to journal bearings. (7)
b. Design a Journal bearing for a steam turbine whose shaft is supported on two bearings one at each side of the turbine and is coupled with a generator for pump production. The weight of the turbine with shaft is 40kN and shaft rotates at 1500 rpm. Diameter of shaft is 100mm. (18)
- OR**
- VI. a. Explain Hydrodynamic theory of lubrication. (5)
b. A shaft rotating at 1440 rpm is supported by two bearings. The forces on each bearings are 6000N and 3500N axial thrust. If shaft diameter is 40mm and expected life of bearing is 500 hours, select a suitable bearing if the required reliability of bearing is 99%. (20)
- VII. a. Explain about general recommendations for the design of welded joints. (12)
b. Explain about modifications of design for manufacturing easiness. (13)
- OR**
- VIII. a. What are the main design recommendations of forged parts? (13)
b. What are the main design recommendation for rolled parts? (12)

